

Quantum Teleportation

Detailed Notes — Quantum Computing

1. What is Quantum Teleportation?

Quantum teleportation is a quantum-information protocol that transfers the quantum state of one qubit to another distant qubit. It does not teleport matter or a physical particle, and it does not allow faster-than-light communication. The protocol uses an unknown quantum state, a shared entangled pair, a Bell-state measurement, two classical bits, and quantum corrections.

The state to be transferred is written as:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

where $|\alpha|^2 + |\beta|^2 = 1$. Alice does not need to know α and β . After the protocol, Bob obtains the same quantum state while Alice's original state has been destroyed by measurement.

2. The Three Qubits

Qubit	Location	Purpose
Qubit 1	Alice	Unknown state $ \psi\rangle$ to be teleported
Qubit 2	Alice	Alice's half of the entangled pair
Qubit 3	Bob	Bob's half of the entangled pair

3. Step 1 — Create an Entangled Pair

Start with two qubits in $|00\rangle$. Apply a Hadamard gate to the first qubit:

$$H|0\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$$

Then apply CNOT with the first qubit as control and the second as target:

$$|00\rangle \rightarrow (|00\rangle + |11\rangle)/\sqrt{2}$$

This is the Bell state $|\Phi\rangle$. Alice keeps one qubit and Bob keeps the other.

$$|\Phi\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$$

4. Complete Initial State

The unknown qubit and Bell pair form the three-qubit state:

$$|\Psi\rangle = |\psi\rangle \otimes |\Phi\rangle$$

Expanding it:

$$|\Psi\rangle = (1/\sqrt{2})[\alpha|000\rangle + \alpha|011\rangle + \beta|100\rangle + \beta|111\rangle]$$

5. Step 2 — Alice Applies CNOT

Alice applies CNOT from the unknown qubit q_1 to her entangled qubit q_2 . The state becomes:

$$|\Psi\rangle = (1/\sqrt{2})[\alpha|00\rangle + \alpha|01\rangle + \beta|11\rangle + \beta|10\rangle]$$

6. Step 3 — Alice Applies Hadamard

Alice applies H to q₁. The resulting state can be rearranged as:

$$|\Psi\rangle = 1/2 [|00\rangle(\alpha|0\rangle + \beta|1\rangle) + |01\rangle(\alpha|1\rangle + \beta|0\rangle) + |10\rangle(\alpha|0\rangle - \beta|1\rangle) + |11\rangle(\alpha|1\rangle - \beta|0\rangle)]$$

This equation is the mathematical heart of quantum teleportation. Alice's two-qubit measurement result determines which transformed version of the original state Bob has.

7. Measurement Results and Corrections

Alice's result	Bob temporarily has	Bob's correction
00	$ \psi\rangle$	I (nothing)
01	$X \psi\rangle$	X
10	$Z \psi\rangle$	Z
11	$XZ \psi\rangle$	X and Z

8. Step 4 — Alice Measures

Alice measures her two qubits. There are four possible outcomes: 00, 01, 10, and 11. Each outcome occurs with probability 1/4 for the standard teleportation setup. The measurement destroys the original unknown quantum state at Alice's side.

Alice then sends the two classical measurement bits to Bob. For example, if her result is 01, Bob learns that his qubit is $X|\psi\rangle$ and must apply X.

9. Step 5 — Bob Applies the Correction

Bob uses Alice's two classical bits to select the required Pauli correction. After applying the correction, Bob's qubit becomes exactly the original state:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

10. Why Entanglement is Necessary

Alice cannot simply measure the unknown state and send the result to Bob. Measurement generally destroys the superposition information. The shared entangled pair provides the quantum resource that allows the state information to be transferred after Alice's Bell-basis measurement.

11. Does Alice Send the Qubit?

No. Alice's original qubit remains physically with Alice until it is measured. Bob's qubit was already physically with Bob as part of the entangled pair. What is transferred is the quantum state, not the physical particle itself.

12. Does Quantum Teleportation Break the Speed of Light?

No. Bob needs the two classical bits from Alice before he knows which correction to apply. Classical communication cannot be used to transmit information faster than light. Therefore quantum teleportation

does not provide faster-than-light communication.

13. Relation to the No-Cloning Theorem

Teleportation does not create two copies of an unknown quantum state. Alice's original state is destroyed during measurement, while Bob reconstructs the state on his qubit. This is consistent with the quantum no-cloning theorem.

14. Example: Superposition State

Consider the state:

$$|\psi\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$$

Alice can teleport this state without explicitly knowing its amplitudes. After the appropriate correction, Bob's qubit is in the same superposition state.

15. Bloch Sphere Interpretation

A single-qubit pure state can be represented on the Bloch sphere as:

$$|\psi\rangle = \cos(\theta/2)|0\rangle + e^{i\phi} \sin(\theta/2)|1\rangle$$

Teleportation can therefore be visualized as transferring the complete qubit state, including its amplitudes and relative phase, to Bob's qubit. Depending on Alice's measurement result, Bob's state may initially differ by X, Z, or XZ. The correction gates rotate it to the target state.

16. Important Quantum Gates

Gate	Matrix / Action	Role
Hadamard H	$1/\sqrt{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$	Creates and transforms superposition
CNOT	$ a, b\rangle \rightarrow a, a \oplus b\rangle$	Creates entanglement and helps perform Bell measurement
Pauli-X	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	Bit flip: $ 0\rangle \leftrightarrow 1\rangle$
Pauli-Z	$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$	Phase flip: $ 1\rangle \rightarrow - 1\rangle$

17. Compact Mathematical Summary

Initial state:

$$|\psi\rangle|\Phi\rangle$$

After Alice's CNOT and Hadamard:

$$1/2 [|00\rangle|\psi\rangle + |01\rangle X|\psi\rangle + |10\rangle Z|\psi\rangle + |11\rangle XZ|\psi\rangle]$$

Alice measures her two qubits and sends the resulting two classical bits to Bob. Bob applies the corresponding Pauli correction. Finally:

$$|\psi\rangle_{\text{Bob}} = |\psi\rangle_{\text{original}}$$

18. Applications

- Quantum communication and quantum networks

- Quantum repeaters for long-distance quantum communication
- Distributed quantum computing
- Quantum internet architectures
- Quantum error correction and fault-tolerant quantum computing

19. Recommended Visualization for a Quantum Learning Platform

For an interactive learning simulator, quantum teleportation should be visualized as a sequence rather than as a static circuit. Show the unknown state $|\psi\rangle$, its Bloch-sphere representation, the entanglement-generation stage, Alice's Bell measurement, the two classical bits travelling to Bob, Bob's conditional X/Z correction, and the final state. Display the mathematical state vector beside the animation so learners can connect the visual change to the underlying equations.

Suggested flow: $|\psi\rangle \rightarrow$ Bell pair creation $\rightarrow$ CNOT $\rightarrow$ H $\rightarrow$ measurement $\rightarrow$ classical bits $\rightarrow$ conditional correction $\rightarrow |\psi\rangle$ at Bob.

20. Key Takeaways

- Quantum teleportation transfers a quantum state, not matter.
- It requires a shared entangled pair.
- Alice performs a Bell-basis measurement on two qubits.
- Alice sends exactly two classical bits to Bob.
- Bob applies X and/or Z corrections based on those bits.
- The original state is destroyed, so teleportation does not violate no-cloning.
- The protocol does not allow faster-than-light communication.
- The final state at Bob is identical to the original unknown state at Alice.